

CN - Computer Networks

From your notes: *TCP, UDP, Routing*

These 3 are core parts of how data moves across the internet.

1. TCP - Transmission Control Protocol

TCP is a *connection-oriented, reliable* protocol. It works at the Transport Layer, Layer 4 of OSI.

Key features:

- *Connection setup*: Uses 3-way handshake `SYN → SYN-ACK → ACK` before sending data
- *Reliable*: Guarantees delivery. If a packet is lost, it retransmits
- *Ordered*: Packets arrive in the same order they were sent
- *Flow control*: Uses sliding window so sender doesn't overwhelm receiver
- *Error checking*: Checksums to detect corrupted data

Used for: Web browsing HTTP/HTTPS, Email SMTP, File transfer FTP.

Analogy: Like calling someone on phone. You make sure they picked up before you start talking.

Downside: Slower due to overhead, but very reliable.

2. UDP - User Datagram Protocol

UDP is a *connectionless, unreliable* protocol. Also Transport Layer.

Key features:

- *No connection*: Just sends packets and forgets. No handshake
- *No guarantee*: Packets may be lost, duplicated, or arrive out of order

- *No flow control*: Sends as fast as it can
- *Low overhead*: Header is only 8 bytes vs TCP's 20 bytes

Used for: Video streaming, Online gaming, DNS, VoIP, Live broadcasts.

Analogy: Like sending a postcard. Fast, but you don't know if it arrived.

Downside: Not reliable, but great for speed and real-time apps.

TCP vs UDP quick compare:

| **TCP** | **UDP**

Connection | Yes | No

Reliability | Reliable | Not reliable

Speed | Slower | Faster

Use case | Web, Email | Video, Gaming, DNS

3. Routing

Routing is the process of *finding the best path* for data to travel from source to destination across networks.

Key points:

- Happens at *Network Layer, Layer 3*
- Done by *Routers* using *Routing Tables*
- A packet may go through many routers before reaching you

Types of Routing:

1. *Static Routing*: Admin manually sets paths. Good for small networks, no overhead
2. *Dynamic Routing*: Routers talk to each other and update paths automatically

Protocols: RIP, OSPF, BGP

How it works:

1. Router receives packet with destination IP
2. Checks routing table for best next hop
3. Forwards packet
4. Repeat until packet reaches destination

Routing Algorithms:

- *Distance Vector*: Routers share info with neighbors. Ex: RIP
- *Link State*: Routers know full map of network. Ex: OSPF

Relation:

App data → TCP/UDP adds port info → IP adds address → Router uses Routing to deliver it.

So: *TCP/UDP decide HOW data is sent*, *Routing decides WHERE it goes*.

Want me to:

1. *Draw a diagram* showing TCP handshake + routing path?
2. *Give examples* of when to choose TCP vs UDP for your projects?
3. *Make notes + PYQs* for exam from this CN section?